

Section: Electricity

Topic: Monitoring and Measuring a.c.

A student measures the output voltage of a signal generator connected to an oscilloscope.

- The oscilloscope trace shows a waveform with:
 - **Peak voltage:** $V_{\text{peak}} = 5.0 \text{ V}$
 - **Period:** $T = 4.0 \text{ ms} = 4.0 \times 10^{-3} \text{ s}$

(a)

Calculate the frequency of the a.c. voltage.

Use:

$$f = \frac{1}{T} = \frac{1}{4.0 \times 10^{-3}} = 250 \text{ Hz}$$

(b)

Calculate the r.m.s. voltage of the a.c. voltage.

Use:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{5.0}{1.414} = 3.54 \text{ V}$$

(c)

The oscilloscope’s timebase is set to 2.0 ms/div.
Verify this is consistent with the measured period.

The waveform has a period of 4.0 ms.
At 2.0 ms/div, one full wave should span:

$$\frac{4.0 \text{ ms}}{2.0 \text{ ms/div}} = 2 \text{ divisions}$$

✔ This matches the oscilloscope reading → **setting is correct.**

🧠 Revision Tips:

- **r.m.s. voltage** gives the equivalent **DC heating effect**
- For sine waves:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}, \quad f = \frac{1}{T}$$

- Timebase (ms/div) × number of divisions = **period of the wave**